mRNA results in the alteration of amino acid sequence of protein that is specified is called RNA editing. This results in the change in the protein coding sequence of RNA following transcription. The coding properties of the RNA transcript is changed. The genetic information encoded in the chloroplast genome is altered by post transcriptional phenomenon which is site – specific (C → U) in chloroplast of higher plants – RNA editing occurs in plant mitochondria and chloroplast.

In plant cells RNA editing by pyrimidine transitions occurs in mitochondria and plastids (chloroplast). There are two main types of RNA editing. (1) Substitution editing – Alteration of individual nucleotide bases. Mitochondria and chloroplast RNA in plants. (2) Insertion / Deletion editing – Nucleotides are added or deleted from the total number of bases.

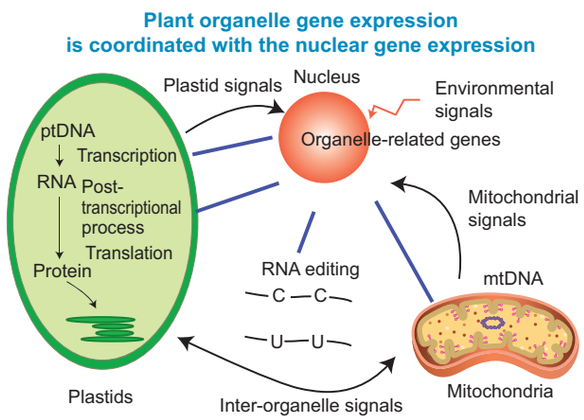


Figure: 3.25 RNA Editing – Post

Significance of RNA Editing

1. In higher plant chloroplast, it helps to restore the codons for conserved amino acids which include initiation and termination codon.
2. It regulates Organellar gene expression in plants.
3. RNA editing results in the restoration of codons for phylogenetically conserved amino acid residues.

3.7.6 Jumping Genes



Figure: 3.26 Barbara McClintock

Have you heard of Jumping Genes or Hopping Genes?

This is the nick name of transposable genetic elements. Transposons are the DNA sequences which can move from one position to another position in a genome. This was first reported by American Geneticist Barbara McClintock as "mobile controlling element" in Maize. One of the most significant scientists of 20th century was Barbara McClintock because she gave a shift in gene organization. McClintock was awarded Nobel prize in 1983 for her work on transposons. Barbara McClintock when studying aleurone of single maize kernels, noted the unstable inheritance of the mosaic pattern of blue, brown and red spots

Year	Editing type	Organelle in Plant cell	Target	Inference / Result
1989	C → U	**Plant mitochondria**	mRNA	For conserved amino acids, multiple changes in codon takes place
1990	U → C	**Plant mitochondria**	mRNA	First report on editing (U → C)
1991	C → U	**Plant chloroplast**	mRNA	First report in **chloroplast**

Table 3.8: RNA Editing types

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